

# ACQUIESCENCE, NOT PERSONALITY: A MULTI-MODEL PSYCHOMETRIC AUDIT OF LLM SELF-REPORT INSTRUMENTS

**Anonymous authors**

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## ABSTRACT

A fast-growing literature scores large language models on validated human personality inventories and feeds the resulting profiles into “silicon-sample” surveys and persona-conditioned multi-agent simulations of opinion dynamics, polarisation, and social behaviour. We ask the prior measurement question: when human instruments are administered to LLMs, do they measure anything at all? We administer a 221-item battery (IPIP-NEO-120, SD3, ZKPQ-50-CC, EPQR-A; 17 domains; 63 reverse-keyed items) to 20 chat-tuned LLMs from 8 vendor families under a Default no-persona instruction and all 16 MBTI personas with  $k=5$  samples per cell at  $T=0.7$  — 375,700 raw API calls. We then run six standard psychometric validity checks plus a persona stress test. Every check fails. The failures share a single mechanism: a shared acquiescent response style with a directional asymmetry — positive forward-minus-reverse agree-rate gap on normal items (IPIP  $\Delta=+0.31$ ) and negative gap on dark-trait items (SD3  $\Delta=-0.38$ ). Cronbach’s  $\alpha$  tracks reverse-item density at Spearman  $\rho=-0.83$  across the 17 domains ( $\rho=-1.0$  on the five IPIP domains); exploratory factor analysis universally collapses the 5 expected Big Five factors into 3 (Tucker’s  $\varphi=0.993$ ); inter-model variance falls to  $<1\%$ ; and 0/320 (*model, persona*) pairs achieve measurement invariance against Default. Persona prompts move profiles by  $>1$  SD per dimension and are 99.7% nearest-centroid recoverable — enough for *coarse* agent contrast — but cross-scale coherence is 0.839 and forward/reverse pair coherence is 0.696: compliance, not measurement validity. Calibrated against synthetic baselines, LLM alphas hug the Random / Pure-Acquiescence floor and stay far from a Trait+Acquiescence simulation ( $\alpha>0.99$ ). The findings impose scope conditions on silicon-sample and persona-agent simulations: coarse persona contrast is reproducible, but trait-grounded simulation of fine-grained psychological mechanisms, conflict dynamics, polarisation, and extreme positions sits on a measurement layer the instruments do not currently support.

## 1 INTRODUCTION

Large language models are increasingly assigned “personalities” by administering human psychometric inventories, and the resulting trait scores are then used as inputs to downstream ML pipelines — silicon-sample surveys (Argyle et al., 2023; Horton, 2023; Aher et al., 2023; Park et al., 2023; Ferreira et al., 2025) and persona-conditioned multi-agent simulations of opinion dynamics, polarisation, and trust (Chuang et al., 2024; Cau et al., 2025; Chuang et al., 2025; Mannekote et al., 2025). A fast-growing literature reports Big Five or Dark Triad scores for chat-tuned LLMs (Miotto et al., 2022; Serapio-García et al., 2025; Jiang et al., 2024; Huang et al., 2023; Bodroža et al., 2024) and treats them as measuring a stable property of the model. We audit the prior evaluation question: do the human instruments measure anything when administered to an LLM?

This promise rests on a prior measurement question that is almost never asked: do the human instruments measure anything when they are administered to an LLM? In the human-subjects tradition, an inventory is not trusted until it passes a battery of psychometric checks — internal consistency, factor structure, convergent validity, and measurement invariance (Cronbach, 1951; Kaiser, 1960; Vandenberg & Lance, 2000). On LLMs these checks are rarely run; when they appear

at all, they typically appear one at a time, on one inventory, on a handful of models (Sühr et al., 2025; Tosato et al., 2026). Larooij & Törnberg (2025) make the same observation about generative agent-based modelling: validation, not capability, is the central bottleneck, and current LLM-driven social simulations rely on face validity rather than measurement-grounded calibration.

This paper runs the missing audit. We administer a 221-item battery combining four validated human inventories — the IPIP-NEO-120 Big Five facets (Johnson, 2014), the Short Dark Triad SD3 (Jones & Paulhus, 2014), the cross-cultural ZKPQ-50-CC (Aluja et al., 2006), and the abbreviated Eysenck EPQR-A (Francis et al., 1992) — to 20 instruction-tuned LLMs from 8 vendor families under a Default no-persona prompt and all 16 MBTI persona prompts, with  $k = 5$  samples per item at temperature 0.7. The completed corpus contains 375,700 raw API calls. We then ran six standard validity checks and a persona stress test. Every check fails. The failures share a single mechanism: a shared acquiescent response style with a directional asymmetry — a positive forward-minus-reverse agree-rate gap on normal-personality items (IPIP  $\Delta = +0.31$ ) and a negative gap on dark-trait items (SD3  $\Delta = -0.38$ ). Acquiescence and social-desirability rejection are not two phenomena (Couch & Keniston, 1960; Crowne & Marlowe, 1960; Paulhus, 1991; Billiet & McClelland, 2000); they are the same acquiescent agree-impulse pointed in opposite directions by item valence.

The downstream cascade is mechanical. Item-level acquiescence inflates inter-item correlations on items it agrees with and depresses them on reverse-keyed items, so Cronbach’s  $\alpha$  tracks reverse-item density at Spearman  $\rho = -1.0$  across the five IPIP Big Five domains. Inflated inter-domain correlations collapse the expected five-factor Big Five structure into three factors, universally across all 20 models, with Tucker’s  $\varphi = 0.993$  and exact replication under leave-one-model-out, per-family, and per-model exploratory factor analysis. By the model level, sequential variance decomposition leaves the Model term at 0.35% of total sum of squares on the Likert composite and 0.51% on the binary composite — cross-model personality comparisons are statistically meaningless at this resolution. Calibration against synthetic baselines on the same item structure makes the diagnosis explicit: LLM observed alphas hug the Random ( $\alpha \approx 0$ ) and Pure-Acquiescence ( $\alpha \approx 0$ ) baselines and stay far below a Trait+Acquiescence simulation ( $\alpha > 0.99$ ). LLMs do not look like a personality with acquiescence on top; they look like acquiescence with no personality underneath.

A persona stress test rules out the obvious rescue — that LLMs only look uniform under a neutral instruction and that personality emerges when the model is asked to play a character. MBTI prompts move profiles by more than one standard deviation per dimension (mean Persona Separation Degree = 1.198) and the resulting profiles are 99.7% nearest-centroid recoverable (319/320 (model, persona) pairs). This is the level of separation that lets persona-conditioned multi-agent simulations differentiate “agent A” from “agent B” (Chuang et al., 2024; Cau et al., 2025). But cross-scale coherence on the four overlapping constructs is only 0.839, forward/reverse pair coherence is 0.696, and the three-factor collapse persists inside every persona condition. Persona steering replaces the acquiescent response style with persona-driven compliance; it does not restore measurement validity. This dissociation between role compliance and trait-grounded behaviour has been hinted at in recent self-report-versus-behaviour comparisons (Han et al., 2025; Mannekote et al., 2025); the present audit gives it a quantitative form and bounds what trait-conditioned LLM agents can defensibly model.

Our contributions are:

- **The first large-scale multi-model, multi-instrument psychometric audit of LLM self-report** — 20 models  $\times$  8 vendor families  $\times$  4 validated instruments  $\times$  17 personas  $\times$   $k=5$  samples = 375,700 API calls, diagnostic rather than profiling.
- **A single-mechanism diagnosis with a directional asymmetry**: acquiescence unifies sycophancy (Perez et al., 2023; Sharma et al., 2024) and safety-driven rejection (Bai et al., 2022) as one agree-impulse flipped by item valence.
- **A falsifiable structural verdict**: removing acquiescence (within-respondent centering) recovers raw-score  $\alpha$  but not the universal 5 $\rightarrow$ 3 factor collapse; synthetic-baseline calibration fits the data to Pure-Acquiescence over Trait+Acquiescence ( $\alpha > 0.99$ ).
- **A persona stress test** showing coarse role compliance without measurement validity (PSD 1.198, 99.7% nearest-centroid recovery, cross-scale coherence 0.839, Fwd/Rev coherence 0.696).
- **A reusable six-check go/no-go diagnostic pipeline** and scope conditions for silicon-sample and persona-conditioned-agent ML applications.

The six-check battery and the persona stress test that organise the rest of the paper are summarised schematically in Appendix A (Figure 4).

## 2 RELATED WORK

### 2.1 LLM PERSONALITY PROFILING, SILICON SAMPLES, AND OUR AUDIT’S PLACE

A dominant strand administers a human inventory to one or more LLMs and reports a profile — NEO-PI on GPT-3 (Miotto et al., 2022), a psychometric framework on PaLM (Serapio-García et al., 2025), PersonaLLM (Jiang et al., 2024), the PsychoBench suite (Huang et al., 2023), temporal-stability tests (Bodroža et al., 2024), and the TRAIT testset (Lee et al., 2025) — with adjacent cognitive-probe (Binz & Schulz, 2023) and trait-descriptor (Suh et al., 2024) work. A parallel strand turns these profiles into operational predictions: silicon-sample surveys (Argyle et al., 2023; Horton, 2023; Aher et al., 2023; Park et al., 2023; Ferreira et al., 2025) and persona-conditioned agents in social simulation (Chuang et al., 2024; Cau et al., 2025; Chuang et al., 2025), which inherit whatever measurement error sits in the underlying scores (Tjautja et al., 2024; Salecha et al., 2024; Larooij & Törnberg, 2025). A growing critical line documents social-desirability bias (Salecha et al., 2024), validity challenges (Sühr et al., 2025), instability (Tosato et al., 2026), self-report/behaviour dissociation (Han et al., 2025; Mannekote et al., 2025), and psychometric-difference benchmarks (Xie et al., 2025; Dong et al., 2025). Almost all of this work uses a single instrument, a single check, and a handful of models.

Our audit differs from the two most related studies. Sühr et al. (2025) show measurement invariance fails for BFI-2 on three GPT-family models; we extend this by (i) naming acquiescence as the upstream mechanism, (ii) quantifying its directional asymmetry, (iii) showing persona steering does not repair the instruments, and (iv) scaling to 20 models and 4 instruments. Serapio-García et al. (2025) reach the opposite conclusion, but their BFI-2 carries  $\approx 15\%$  reverse-keyed items (versus 29% here), they sample at  $T=0$ , and they do not test reverse-item consistency — the central failure we identify. More broadly, sycophancy, social-desirability, and prompt sensitivity have each been documented as separate behaviours (Perez et al., 2023; Sharma et al., 2024; Salecha et al., 2024); our claim is a measurement-level verdict about their joint, quantifiable, and (under persona steering) non-repairable damage to instrument validity.

### 2.2 RESPONSE STYLES, ACQUIESCENCE, AND SOCIAL DESIRABILITY

Six decades of survey methodology catalogue response styles — acquiescence (Couch & Keniston, 1960), social desirability (Crowne & Marlowe, 1960; Paulhus, 1991), and their structural-equation treatment (Billiet & McClendon, 2000; Baumgartner & Steenkamp, 2007) — with reverse-keyed items as the standard countermeasure. Our audit imports the classical toolkit wholesale: Cronbach’s  $\alpha$  (Cronbach, 1951), classical test theory (Lord & Novick, 1968), EFA with Kaiser (Kaiser, 1960), parallel analysis (Horn, 1965) and varimax (Kaiser, 1958), Tucker’s congruence (Tucker, 1951), measurement invariance (Meredith, 1993; Vandenberg & Lance, 2000), forced-choice/IRT alternatives (Brown & Maydeu-Olivares, 2011; Li et al., 2025), and bootstrap (Efron, 1979) / Mahalanobis (Mahalanobis, 1936) inference.

Two gaps remain in the LLM literature. First, the response-style toolkit has not been applied coherently: where reverse-keyed inconsistency is reported (Braun, 2025), it is a curiosity rather than a falsifiable diagnostic with a structural signature. Second, the directional asymmetry between normal- and dark-trait items is undocumented at the item level. We argue both are symptoms of one acquiescent mechanism, observable through the same forward/reverse decomposition.

### 2.3 ALIGNMENT, RLHF, AND BEHAVIOURAL SIDE EFFECTS

The behavioural side effects of instruction tuning, RLHF, and DPO (Wei et al., 2022; Ouyang et al., 2022; Christiano et al., 2017; Bai et al., 2022; Rafailov et al., 2023) — sycophancy (Perez et al., 2023; Sharma et al., 2024), reduced output diversity (Kirk et al., 2023), agreeableness in role-play (Shah et al., 2026), alignment limits (Barnhart et al., 2025), survey response biases (Tjautja et al., 2024; Domínguez-Olmedo et al., 2024), and self-report/behaviour dissociation (Han et al., 2025) — are well documented as separate behaviours. This strand has not connected them to classical response-

| Instrument                      | Items      | Format           | Doms      | Rev       |
|---------------------------------|------------|------------------|-----------|-----------|
| IPIP-NEO-120 (Johnson, 2014)    | 120        | Likert-5         | 5         | 41        |
| SD3 (Jones & Paulhus, 2014)     | 27         | Likert-5         | 3         | 5         |
| ZKPQ-50-CC (Aluja et al., 2006) | 50         | T / F            | 5         | 12        |
| EPQR-A (Francis et al., 1992)   | 24         | Y / N            | 4         | 5         |
| <b>Total</b>                    | <b>221</b> | <b>3 formats</b> | <b>17</b> | <b>63</b> |

Table 1: The 221-item battery: 4 inventories, 3 response formats, 17 domains, 63 reverse-keyed items (28.5%).

style theory or shown a single, mechanistically traceable failure of measurement validity across the standard checks. We treat acquiescence and its directional inversion on dark-trait items as that unifying mechanism, accounting for the  $\alpha$ -reverse-density relation, the universal 5 $\rightarrow$ 3 collapse (Tucker’s  $\varphi=0.993$ ), inter-model variance below 1%, the item-plasticity dichotomy, and persona compliance without validity (Salecha et al., 2024; Laverghetta et al., 2022).

### 3 METHODOLOGY

**Battery (Table 1).** IPIP-NEO-120 covers the Big Five at the domain level; SD3 adds the three Dark Triad constructs; ZKPQ-50-CC adds Activity, Aggression-Hostility, Impulsive Sensation Seeking, Neuroticism-Anxiety, and Sociability in true/false format; EPQR-A adds Eysenck Psychoticism, Extraversion, Neuroticism, and a Lie scale in yes/no format. The multi-instrument design is required by the convergent-validity and cross-scale coherence checks: no single inventory provides the redundant cross-format measurements those tests need.

**Models and protocol.** We query 20 instruction-tuned LLMs across 8 vendor families: Anthropic Claude-Opus-4.6 and Claude-Sonnet-4.6; OpenAI GPT-5.2 and GPT-5.5 (OpenAI, 2023); Google Gemini-3-Pro / -Flash and Gemini-3.1-Pro / -Flash-Lite (Gemini Team, Google, 2024); DeepSeek V3.2, V4-Flash, V4-Pro (DeepSeek-AI, 2024); Alibaba Qwen3-235B-A22B, Qwen3.5-122B-A10B, Qwen3.5-397B-A17B (Qwen Team, 2024); Zhipu GLM-4.6V, GLM-4.7, GLM-5.1 (GLM Team, Zhipu AI, 2024); Moonshot Kimi-K2.5 and Kimi-K2.6 (Moonshot AI, 2025); and MiniMax-M2.7 (MiniMax, 2025). Each model is queried at  $T=0.7$  with  $k=5$  samples per (model, persona, item) cell. The 17 conditions are a Default no-persona instruction (“Please answer the following personality questionnaire honestly.”) plus all 16 MBTI personas administered as “Respond as if you have <TYPE> personality.”. The completed corpus contains  $20 \times 17 \times 221 = 75,140$  averaged item scores from 375,700 raw API calls; 61 refusals (0.0162%, all under Default) are mean-imputed per item (Appendix C).

**Six validity checks.** (i) *Internal consistency*: Cronbach’s  $\alpha$  (Cronbach, 1951) per IPIP domain over the  $17 \times 20 = 340$  (persona, model) cells. (ii) *Factor structure*: EFA on the  $340 \times 17$  domain matrix (Kaiser (Kaiser, 1960) and parallel-analysis (Horn, 1965) retention, varimax (Kaiser, 1958); leave-one-model-out, per-family, and per-model replication; Tucker’s  $\varphi$  (Tucker, 1951)). (iii) *Reverse-item consistency*: Pairwise Inconsistency Rate and the forward–reverse agree-rate gap  $\Delta$ . (iv) *Variance decomposition*: marginal Type-III SS partition over Model+Domain+Persona+Item+Residual (factors are non-orthogonal, so components need not sum to the total). (v) *Convergent validity*: Pearson/Spearman over 8 pre-specified construct pairs. (vi) *Measurement invariance*: Default-vs-MBTI item-vector Pearson  $r$ , strong-invariance threshold  $r > 0.8$  (Vandenberg & Lance, 2000; Meredith, 1993).

**Persona stress test and baselines.** Persona Separation Degree (PSD) is the normalised 17-d Euclidean distance from Default in z-space divided by  $\sqrt{17}$ . Nearest-centroid persona classification uses leave-one-model-out centroids in z-Euclidean, cosine, and Mahalanobis (Mahalanobis, 1936) metrics. Target adherence, sign-aligned cross-scale coherence, forward/reverse pair coherence, item-level plasticity, and a factorial main-effects model for the four MBTI axes complete the stress test. For calibration, three synthetic baselines on the IPIP structure — Random, Pure Acquiescence, Trait+Acquiescence — anchor floor and ceiling; aggregates carry bootstrap 95% CIs (Efron, 1979).

| Domain            | LLM $\alpha$ | LLM SD | Human $\alpha$ | Gap    | Rev. % |
|-------------------|--------------|--------|----------------|--------|--------|
| Neuroticism       | 0.813        | 0.035  | 0.90           | -0.087 | 17     |
| Extraversion      | 0.928        | 0.008  | 0.89           | +0.038 | 17     |
| Openness          | 0.189        | 0.160  | 0.87           | -0.681 | 25     |
| Agreeableness     | 0.069        | 0.311  | 0.88           | -0.811 | 71     |
| Conscientiousness | 0.535        | 0.113  | 0.90           | -0.365 | 25     |

Table 2: Cronbach’s  $\alpha$  per IPIP Big Five domain, averaged across 20 LLMs. Human baseline from Johnson (2014). The gap to the human baseline is monotonically increasing in reverse-keyed item percentage; the Spearman rank correlation between LLM  $\alpha$  and reverse percentage across the 5 domains is exactly  $-1.0$  ( $p < 0.01$ ).

## 4 EXPERIMENTS AND RESULTS

### 4.1 INTERNAL CONSISTENCY: $\alpha$ TRACKS REVERSE DENSITY AT $\rho = -1.0$

Cronbach’s  $\alpha$  per IPIP Big Five domain, averaged across the 20 models (Table 2), reveals a clean rank-1 inverse relationship with reverse-keyed item density: Extraversion (17% reverse,  $\alpha = 0.928$ ), Neuroticism (17%,  $\alpha = 0.813$ ), Conscientiousness (25%,  $\alpha = 0.535$ ), Openness (25%,  $\alpha = 0.189$ ), Agreeableness (71%,  $\alpha = 0.069$ ). Spearman  $\rho(\alpha, \text{reverse}\%) = -1.0$  ( $p < 0.01$ ); Figure 1 plots the gradient. Notice that two domains (Extraversion and Neuroticism) actually equal or exceed the human baseline — this is not evidence of strong personality measurement on LLMs but a side effect of low reverse-item density: items the model is willing to agree with all cohere because the agree-impulse drives them in the same direction. The inverse relationship is not an artefact of the five IPIP domains: across all 17 domains the Spearman correlation between raw-score  $\alpha$  and reverse-item percentage is  $\rho = -0.83$  ( $p < 0.001$ ;  $\rho = -0.75$  on the 14 non-degenerate domains), and the IPIP-five  $\rho = -1.0$  is simply the clean end of a general law. We emphasise that this raw-score  $\alpha$  pattern is a *diagnostic signature of acquiescence* (forward and reverse items endorsed alike, destroying raw-score consistency in proportion to reverse density), not a claim that the items are unreliable; §4.4 confirms this directly by removing acquiescence and showing that the signature is removable while the structural failure is not.

### 4.2 ITEM-LEVEL ACQUIESCENCE WITH DIRECTIONAL ASYMMETRY

Decomposing every model’s responses into forward and reverse agree-rates per instrument (Table 4) reveals two opposite signs of the same bias: IPIP-NEO-120 forward-vs-reverse  $\Delta = +0.31$  (classic acquiescence on normal-personality items); SD3  $\Delta = -0.38$ , ZKPQ-50-CC  $\Delta = -0.34$ , EPQR-A  $\Delta = -0.08$  (social-desirability rejection on dark-trait or non-normative items). Aggregated across instruments the gap is near zero ( $+0.03$ ), but at the item level the same response pattern simultaneously produces agreement on socially desirable content and refusal on socially undesirable content (Figure 2). Overall PIR = 0.468 with bootstrap 95% CI [0.413, 0.531]; per-model correlation between reverse-agree rate and PIR is  $+0.63$  (models that acquiesce more are more inconsistent). Persona prompts reduce PIR substantially: Default PIR-agreement of 0.674 vs MBTI mean of 0.834 (paired  $t = 10.70$ ,  $p < 0.0001$ ; Appendix A, Figure 15).

### 4.3 FACTOR STRUCTURE: 5 EXPECTED, 3 OBSERVED, UNIVERSALLY

Both Kaiser ( $\lambda > 1$ ) and parallel analysis retain exactly **3 factors** on the  $340 \times 17$  matrix, against the expected 5 from Big Five theory (eigenvalues  $\lambda_1 = 6.97$ ,  $\lambda_2 = 4.33$ ,  $\lambda_3 = 3.07$ ; first three factors explain 86–90% of variance). The varimax loadings (Figure 3, full numerical table in Appendix A, Table 5) show that (i) Extraversion-flavoured constructs — IPIP Extraversion, ZKPQ Activity and Sociability, EPQR Extraversion, SD3 Narcissism — collapse into a single F1; (ii) Neuroticism merges with (inverted) Agreeableness and Machiavellianism into F2; and (iii) Conscientiousness, EPQR Lie, and (inverted) ZKPQ Impulsive Sensation Seeking and EPQR Psychoticism load on F3. The collapse is universal: 20 leave-one-model-out reruns all return exactly 3 Kaiser factors (first-eigenvalue range 6.94–7.02; Tucker’s  $\varphi = 0.993$ ); 8 per-family reruns and 20 per-model reruns all return exactly 3 Kaiser factors (Appendix A, Figure 13).

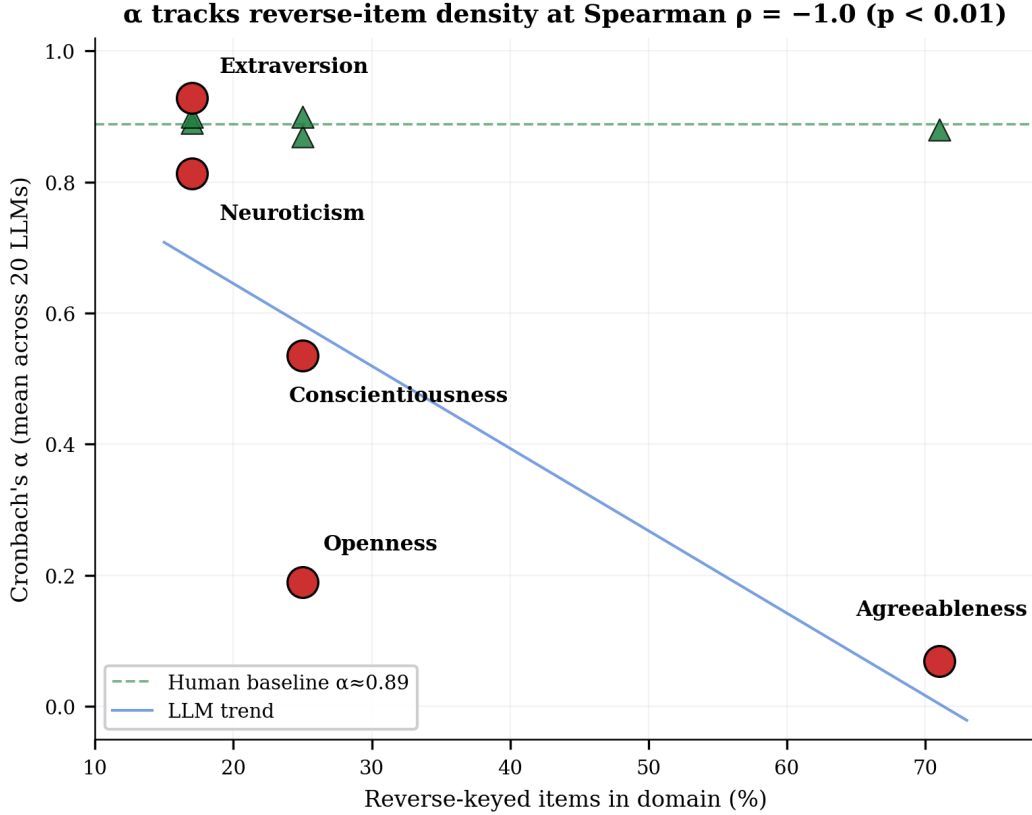


Figure 1: Cronbach’s  $\alpha$  (mean across 20 LLMs) vs reverse-keyed item percentage per IPIP Big Five domain. The 5 domains line up exactly along an inverse trend (Spearman  $\rho = -1.0$ ,  $p < 0.01$ ). Human baseline  $\alpha \approx 0.88$  shown for reference. Reverse-keyed items expose the acquiescent response style.

#### 4.4 ACQUIESCENCE CORRECTION: SYMPTOM VERSUS STRUCTURE

A reader may worry that the low raw-score  $\alpha$  in reverse-heavy domains (§4.1) is merely an artefact of computing  $\alpha$  on un-reverse-scored items, rather than evidence of acquiescence. We test this directly in two steps. First, reverse-scoring the reverse-keyed items before computing  $\alpha$  restores high  $\alpha$  on every IPIP domain (mean 0.94 versus 0.51 on raw scores), confirming that the  $\alpha$ -versus-reverse-density pattern of Figure 1 is a *diagnostic signature of acquiescence* (forward and reverse items endorsed alike) rather than a claim that the items are unreliable. Second, and more decisively, we apply the textbook statistical correction for acquiescence — within-respondent mean-centering across all items (Couch & Keniston, 1960) — and re-run the EFA on the 17-domain matrix. The correction does *not* recover the Big Five: the structure still yields three factors under both the Kaiser criterion and parallel analysis, with the eigenvalue profile essentially unchanged (top three 6.97/4.33/3.07  $\rightarrow$  7.00/3.73/3.46; the fourth eigenvalue remains below 1 in both cases). If a coherent five-factor structure were merely masked by acquiescence, removing it should reveal more factors; it does not. Two conclusions follow. The surface symptom (raw-score  $\alpha$  tracking reverse density) and the structural verdict (five $\rightarrow$ three collapse) are decoupled: the former is removable by reverse-scoring, the latter is not removable by the standard correction. And because high  $\alpha$  after correction can reflect shared method variance rather than a recovered trait,  $\alpha$  alone is not evidence of validity (Cronbach, 1951); the factor-structural collapse is the load-bearing result. This is consistent with the synthetic-baseline calibration below: the data resemble acquiescence-with-no-trait, not trait-plus-acquiescence.

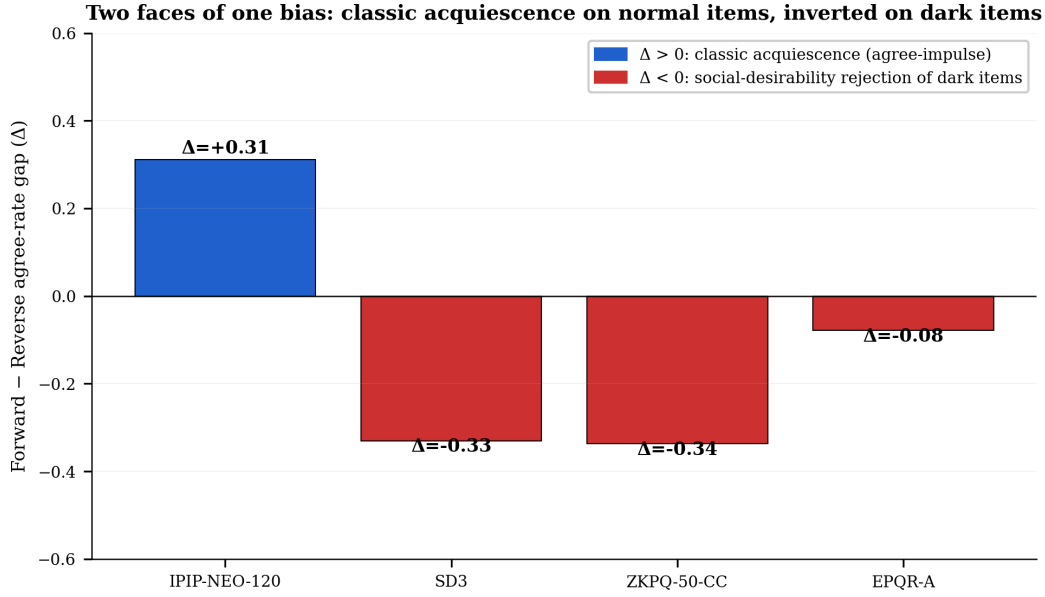


Figure 2: Forward-minus-reverse agree-rate gap per instrument, averaged across 20 LLMs. Positive bar (IPIP) is classic acquiescence; negative bars (SD3, ZKPQ, EPQR-A) are social-desirability rejection. Two faces of the same acquiescent agree-impulse.

| Source       | Likert (%)  | Binary (%)  |
|--------------|-------------|-------------|
| <b>Model</b> | <b>0.35</b> | <b>0.51</b> |
| Domain       | 23.93       | 5.27        |
| Persona      | 4.09        | 14.06       |
| Item         | 36.13       | 14.81       |
| Residual     | 35.51       | 65.35       |

Table 3: Marginal variance decomposition (% of total Sum-of-Squares) into Model + Domain + Persona + Item + Residual terms, computed separately for the Likert (IPIP+SD3) and Binary (ZKPQ+EPQR) composites. Marginal components need not sum to 100%. Inter-model variance is below 1% on both composites (and remains below 2% even under a sequential Type-I partition entering Model first).

#### 4.5 MODEL-LEVEL CONVERGENCE: INTER-MODEL VARIANCE BELOW 1%

Marginal variance decomposition (Table 3; Appendix A, Figure 12) shows that inter-model variance accounts for 0.35% of total Sum-of-Squares on the Likert composite and 0.51% on the binary composite. After accounting for the inventory’s own structure (Domain), the prompted persona (Persona), and per-item variation (Item), almost nothing is left for the model. The question “which LLM is more conscientious?” is, at the resolution of these instruments, statistically meaningless — cross-model comparisons are at best ranking models on residual noise.

#### 4.6 CONVERGENT VALIDITY AND CALIBRATION AGAINST SYNTHETIC BASELINES

Across the 20 model-level means we test 8 pre-specified construct pairs (Appendix B, Table 6). Sign-match is 7/8; the single failure is Extraversion  $\times$  ZKPQ Sociability ( $r_p = -0.10$ ,  $p = 0.670$ ). Strongest convergence: Agreeableness  $\times$  Psychopathy ( $r_p = -0.81$ ), Conscientiousness  $\times$  Psychopathy ( $r_p = -0.73$ ), Agreeableness  $\times$  Machiavellianism ( $r_p = -0.71$ ). DIF across Western and Chinese vendor families is non-significant in every domain (all  $p > 0.16$ , max  $|d| = 0.42$ ). To distinguish “trait + noise” from “no trait at all”, we calibrate against three synthetic baselines on the same IPIP structure (Appendix B, Table 7, Figure 17). Random-uniform and Pure-Acquiescence generators both yield  $\alpha \approx 0$ , whereas a Trait+Acquiescence generator (a genuine 5-factor latent

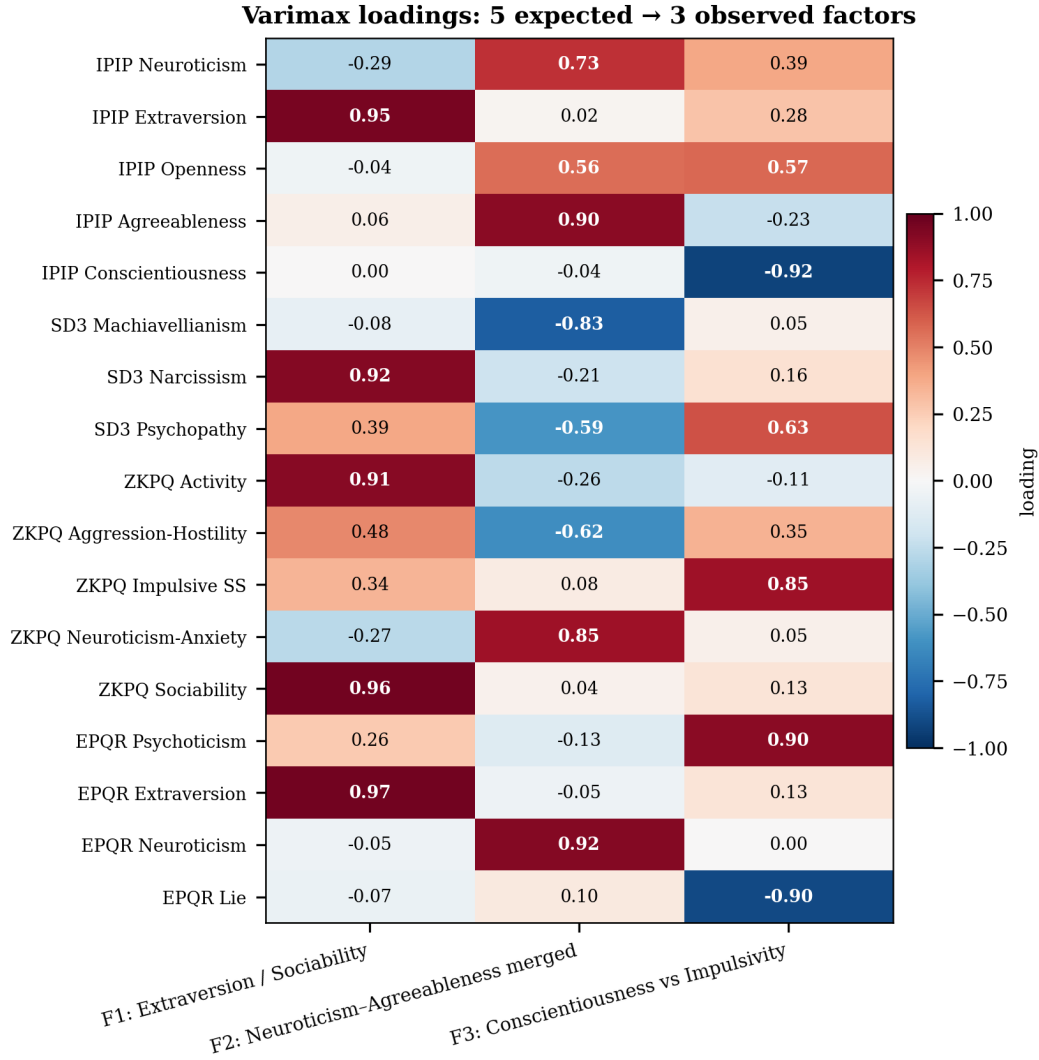


Figure 3: Varimax-rotated factor loadings on the standardised 17-domain matrix ( $n = 340$ ). The expected 5-factor Big Five collapses into 3: Extraversion/Sociability, a fused Neuroticism-Agreeableness, and Conscientiousness vs Impulsivity.

structure plus an acquiescence offset) recovers  $\alpha > 0.99$  on every domain. LLM observed  $\alpha$  sits close to Random / Pure-Acquiescence on the high-reverse-density domains (Agreeableness 0.07, Openness 0.19) and only approaches the human band on low-reverse-density domains where acquiescence inflation cannot be detected (Extraversion 0.93, Neuroticism 0.81). Pure acquiescence with no underlying trait is a quantitatively better generative model of the LLM data than “trait with acquiescence on top.”

#### 4.7 PERSONA STRESS TEST: COMPLIANCE WITHOUT VALIDITY

A persona stress test rules out the rescue that personality emerges only when the model is asked to play a character. MBTI prompts move profiles by more than one SD per dimension (mean PSD = 1.198, per-model range 0.996–1.384, max 1.67; Appendix A, Figure 7). Nearest-centroid persona classification recovers **319 / 320** pairs in z-Euclidean (99.7%; Figure 5); the single failure is DeepSeek-V3.2 ESFP → ENTP. Likert-only reaches 100%, cosine 99.7%, Mahalanobis 98.4%. Empirical target adherence averages 0.838; theoretical-target adherence averages only 0.557 — LLMs reproduce an empirical persona pattern that diverges from the textbook MBTI description.

But persona compliance is not measurement validity. Three statistics fail simultaneously. **Measurement invariance** (Pearson  $r$  between each model’s Default vector and each MBTI vector) averages  $r = 0.461$  across 320 pairs, with 0/320 crossing  $r > 0.8$ . **Cross-scale coherence** of persona-induced shifts across instruments measuring the same construct is 0.839 (95% CI [0.815, 0.862]) — below the 0.997 nearest-centroid fidelity. **Forward/reverse pair coherence** is only 0.696 (0.737 EPQR-A, 0.727 ZKPQ, 0.685 IPIP, 0.657 SD3); on  $\approx 1/3$  of Fwd/Rev pairs the persona shift fails directional sanity. The factorial model also leaks across axes: EPQR Lie loads on J/P ( $\beta = 0.78$ ); SD3 Narcissism loads on E/I ( $\beta = 0.94$ ).

Item-level plasticity (Figure 6) shows a sharp dichotomy: most-plastic items shift up to 0.63 of scale range (EPQR “Do you sometimes talk about things you know nothing about?” 0.632); most-locked items do not move (EPQR “Do you enjoy hurting people you love?” 0.000; “Take advantage of others” 0.056). The plasticity floor is the alignment safety filter, not the persona’s plausible behaviour. Default-condition within-sample SD (0.205 Likert) exceeds every MBTI persona (0.081–0.118): persona prompts *stabilise* responses (Appendix D).

## 5 DISCUSSION: IMPLICATIONS FOR LLM-BASED SOCIAL SIMULATION

A single acquiescent response style with a directional asymmetry ( $\Delta = +0.31$  on normal items;  $\Delta = -0.38$  on dark-trait items) explains why every standard psychometric check fails: raw-score  $\alpha$  tracks reverse density ( $\rho = -0.83$  across 17 domains), EFA collapses 5 Big Five factors into 3 universally (Tucker’s  $\varphi = 0.993$ ), inter-model variance falls below 1%, and 0/320 (*model, persona*) pairs cross the strong-invariance threshold. Removing acquiescence recovers  $\alpha$  but not the factor structure, and calibration places the data near Pure-Acquiescence, far from Trait+Acquiescence — acquiescence-with-no-trait, not trait-plus-style. Persona prompts are 99.7% recoverable yet yield only 0.839 cross-scale and 0.696 Fwd/Rev coherence: coarse role compliance, not measurement validity.

These findings impose scope conditions on the silicon-sample (Argyle et al., 2023; Horton, 2023; Ferreira et al., 2025) and persona-conditioned-agent (Chuang et al., 2024; Cau et al., 2025; Chuang et al., 2025; Park et al., 2023) programmes. Coarse, recoverable persona contrast (99.7% of cells in the right MBTI bin) is enough to differentiate “agent A” from “agent B” and drive demographic stratification, but it is not a trait-grounded substrate: persona shifts disagree across instruments (0.839), reverse pairs fail directional sanity in  $\approx 1/3$  of cases, and the 3-factor collapse persists in every persona condition. Three consequences follow: (1) simulations relying on factor-pure trait differences or trait-mediated causal chains inherit the collapse; (2) simulations of extreme opinions, conflict, or polarisation (Chuang et al., 2025; Cau et al., 2025) hit a safety floor — dark/morality-anchored items have plasticity  $\approx 0$ , so persona agents under-express the very content that drives polarisation; and (3) the implicit ISTJ-like Default (Appendix A, Figure 14) injects a cautious, non-confrontational role into every comparison, so “no-persona” is not neutral. The prescription: report internal consistency, factor structure, convergent validity, and measurement invariance before interpreting any LLM-personality score, and make the “coarse persona contrast” vs “trait-grounded mechanism” choice explicit at design time.

**Future directions.** Because self-report questionnaires are not a valid measurement substrate for current LLMs, we point to four alternatives: (i) *behavioural validation*, inferring traits from downstream choices or interactive tasks rather than self-report (Han et al., 2025; Mannekote et al., 2025); (ii) *forced-choice / ipsative* formats that structurally bypass acquiescence by forcing trade-offs between equally desirable options (Brown & Maydeu-Olivares, 2011; Li et al., 2025); (iii) *item-response-theory* models that separate trait from response-style parameters; and (iv) *behavioural signatures* (consistency patterns, refusal profiles) invariant to prompt framing. To support replication we will release the six-check pipeline and the 375,700-response dataset.

## LIMITATIONS

The study uses closed-weight commercial APIs only; alignment is diagnosed by mechanism rather than direct intervention against base / pre-RLHF baselines. We therefore cannot distinguish whether the shared acquiescent response style originates from alignment, from pre-training data, or from

their interaction, and our causal language is correspondingly circumspect; a preliminary base/instruct comparison (Qwen3-0.6B,  $k=1$ ) gives an overall reverse-item agree rate of 0.70 (base) versus 0.77 (instruct) — both variants are highly acquiescent and instruction tuning shifts the rate only modestly — but a single small model pair is insufficient to localise the source. Instruments and prompts are English; forced-choice, ipsative, behavioural-elicitation, and IRT-fitted alternatives are untested. Temperature is fixed at 0.7 with  $k=5$ ; multi-temperature and multi-seed designs are recommended but not run here. The persona stress test uses only the 16 MBTI types, and the implicit ISTJ-like Default may shift under alternative no-persona phrasings. Our scope conditions on LLM-based social simulation apply at the level of the underlying self-report measurement substrate: we do not evaluate behavioural-trace simulations (Park et al., 2023) or simulations whose outputs are anchored to verifiable external data rather than self-report scores. Conclusions concern currently deployed instruction-tuned chat models and make no claim about future architectures.

## ETHICS STATEMENT

The study involves no human subjects. All data are LLM API responses to published validated inventories used under their original research-use terms. The work is motivated by a public-good concern that unsupported personality-trait claims about LLMs propagate into alignment, hiring, and public-opinion applications. All 61 refusals are reported transparently (Appendix C); no attempt was made to evade safety filters or retry refused calls. Anonymised data and code release are deferred to post-acceptance.

## REPRODUCIBILITY STATEMENT

All 375,700 raw responses were collected via the providers’ official APIs at  $T=0.7$  with  $k=5$  samples per (model, persona, item) cell. The 20 model identifiers and vendor families, the 221-item battery (IPIP-NEO-120, SD3, ZKPQ-50-CC, EPQR-A), the 17 prompt conditions, and the six psychometric checks are fully specified in §3; the analysis pipeline ( $\alpha$ , PIR, EFA with Kaiser/parallel-analysis/varimax, variance decomposition, measurement invariance, persona stress test, synthetic baselines) in §4–5 and Appendices A–E. Upon acceptance we release the response corpus, the analysis code (including the six-check go/no-go pipeline), and the scripts that regenerate every figure and table.

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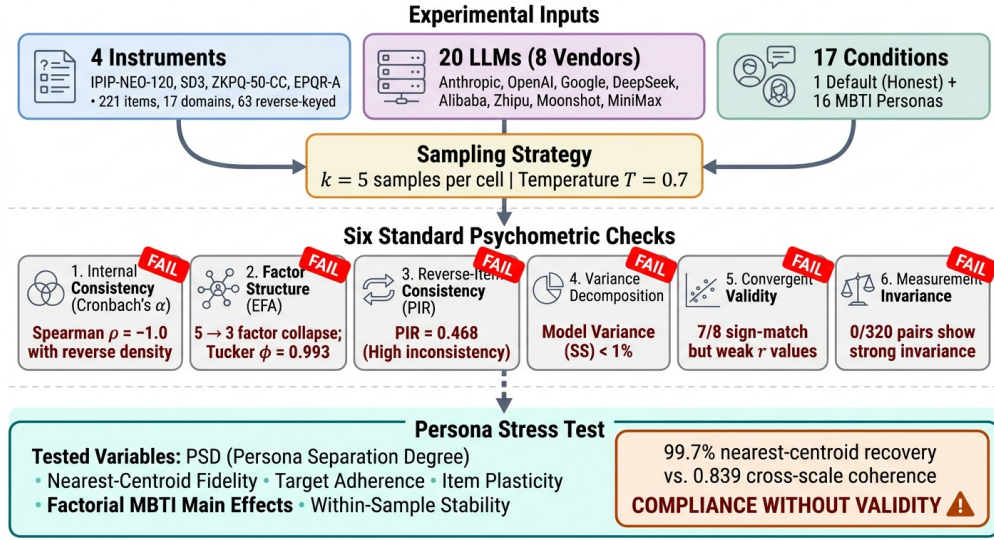


Figure 4: Schematic of the six-check psychometric validation battery plus persona stress test applied to 20 LLMs on the 221-item battery. Each check displays a FAIL badge with its diagnostic outcome.

## A PERSONA STEERING SUPPLEMENT

| Instrument   | Fwd agree | Rev agree | $\Delta$ (Fwd-Rev) |
|--------------|-----------|-----------|--------------------|
| IPIP-NEO-120 | 0.67      | 0.35      | +0.31              |
| SD3          | 0.32      | 0.70      | -0.38              |
| ZKPQ-50-CC   | 0.12      | 0.45      | -0.34              |
| EPQR-A       | 0.08      | 0.16      | -0.08              |
| Overall      | 0.42      | 0.39      | +0.03              |

Table 4: Per-instrument forward (Fwd) and reverse (Rev) item agreement rates and their gap  $\Delta = \text{Fwd} - \text{Rev}$ , Default condition, mean over 20 models (agree = Likert  $\geq 3$  or binary True/Yes). Positive  $\Delta$  on IPIP is classic acquiescence; negative  $\Delta$  on the dark/non-normative instruments is social-desirability rejection. Full per-model table in `results/table4.fwd.rev.delta.csv`.

### A.1 PERSONA FIDELITY AND ITEM PLASTICITY

Figure 5 reports the  $16 \times 16$  confusion matrix of nearest-centroid persona classification with LOO centroids; only one off-diagonal cell. Figure 6 shows the per-item plasticity distribution across content categories.

### A.2 PER-MODEL PERSONA SEPARATION DEGREE AND WITHIN-SAMPLE STABILITY

Figure 7 reports the per-model PSD across the 16 MBTI personas. Mean PSD across all models is 1.198; Gemini-3-Pro is the most steerable at 1.384 and MiniMax-M2.7 the least at 0.996. The peak persona is overwhelmingly ENTP or ESTP — the two MBTI types that push hardest against the implicit ISTJ-like Default baseline.

### A.3 MEASUREMENT INVARIANCE DISTRIBUTION

Figure 8 shows the histogram of Pearson  $r$  between Default and each MBTI item vector across the 320 (model, persona) pairs. Mean  $r = 0.461$ ; the strong-invariance threshold  $r > 0.8$  is crossed by 0/320 pairs.

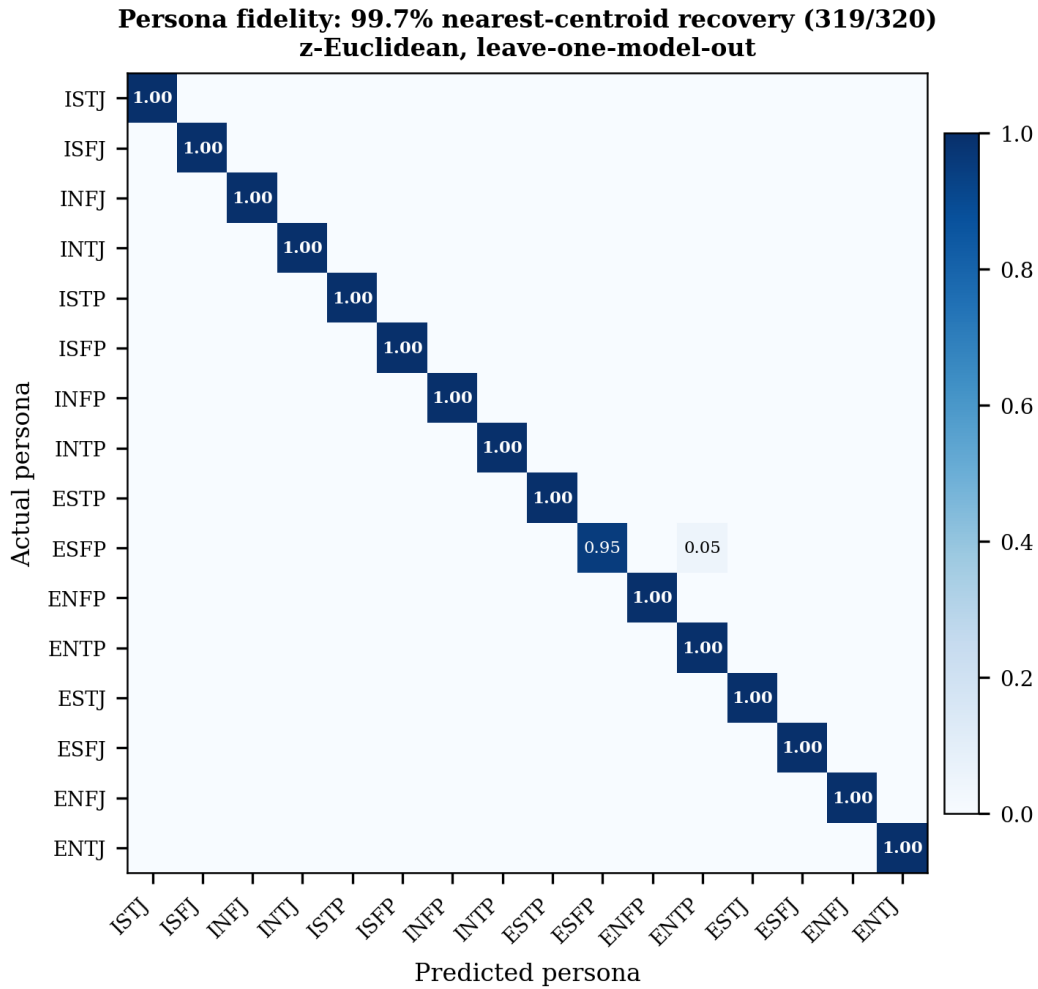


Figure 5: Nearest-centroid persona classification (z-Euclidean, LOO): 319/320 (model, persona) pairs recovered; single confusion is DeepSeek-V3.2 ESFP  $\rightarrow$  ENTP.

#### A.4 CROSS-SCALE COHERENCE AND TARGET ADHERENCE

Figure 9 reports sign-aligned cross-scale coherence by construct (Extraversion 0.898, Neuroticism 0.868, Conscientiousness-vs-Disinhibition 0.800, Agreeableness-vs-Antagonism 0.790; overall 0.839 [bootstrap 95% CI 0.815, 0.862]; sign-match 0.788). The dashed reference at 0.997 marks nearest-centroid persona recovery, exposing the compliance / validity gap.

Figure 10 reports per-persona mean target adherence against the LOO empirical centroid and the hand-coded theoretical MBTI target. Empirical adherence averages 0.838; theoretical adherence averages 0.557 — prompted MBTI personas reproduce an empirical pattern that diverges from the textbook descriptions.

#### A.5 FACTORIAL MBTI MAIN EFFECTS

Figure 11 shows standardised main-effect  $\beta$  for the four MBTI axes across the 17 domain scores. Notable cross-axis leakage: EPQR Lie loads heavily on J/P ( $\beta = 0.78$ ); SD3 Narcissism loads heavily on E/I ( $\beta = 0.94$ ).

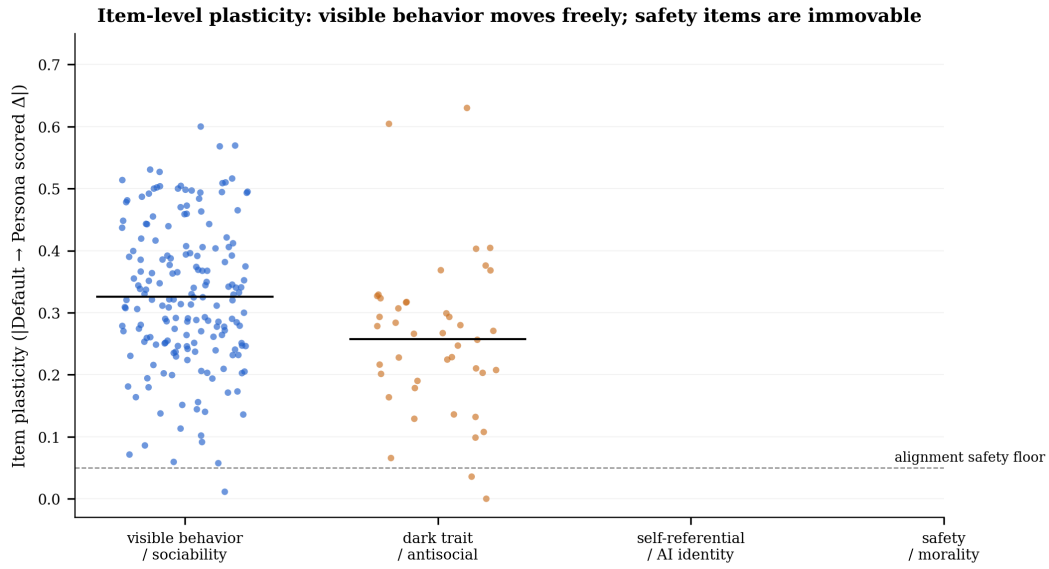


Figure 6: Item-level plasticity (mean absolute Default-vs-persona scored shift, normalised by scale range) across 221 items, by content category. Visible-behaviour items reach 0.6 plasticity; safety / morality items are pinned at 0.

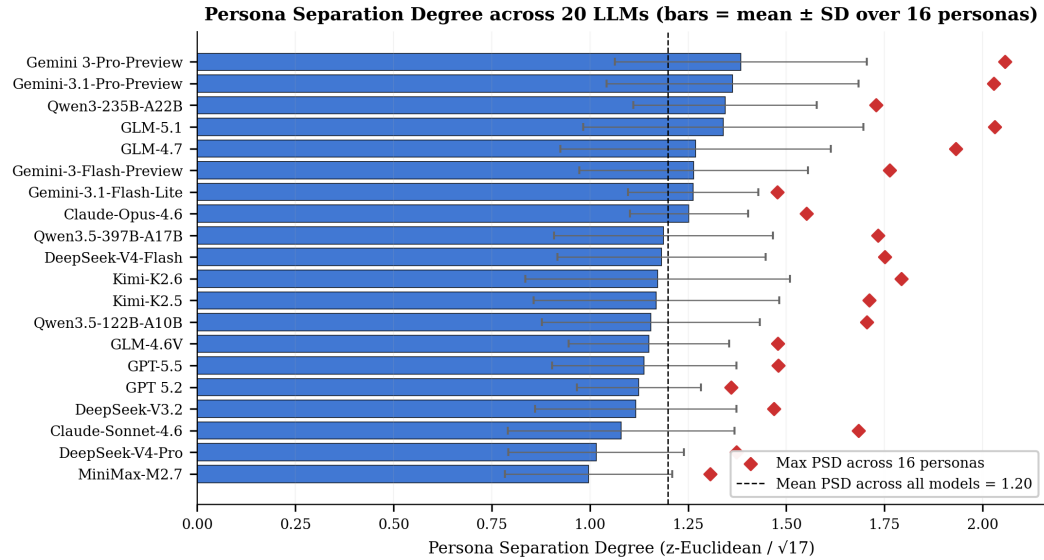


Figure 7: Persona Separation Degree (mean across 16 MBTI personas, bars; max, diamonds; SD as error bars) per model.

#### A.6 EFA ROBUSTNESS AND FULL LOADINGS

Figure 13 overlays the first-eigenvalue distribution from 20 leave-one-model-out reruns on the full-sample eigenvalue spectrum. All 20 LOO reruns return exactly 3 Kaiser factors.

#### A.7 DEFAULT CONDITION BASELINE

Figure 14 shows the nearest-centroid (z-Euclidean) and MBTI-axis classifications of the 20 LLMs under Default. Most models project as ISTJ.

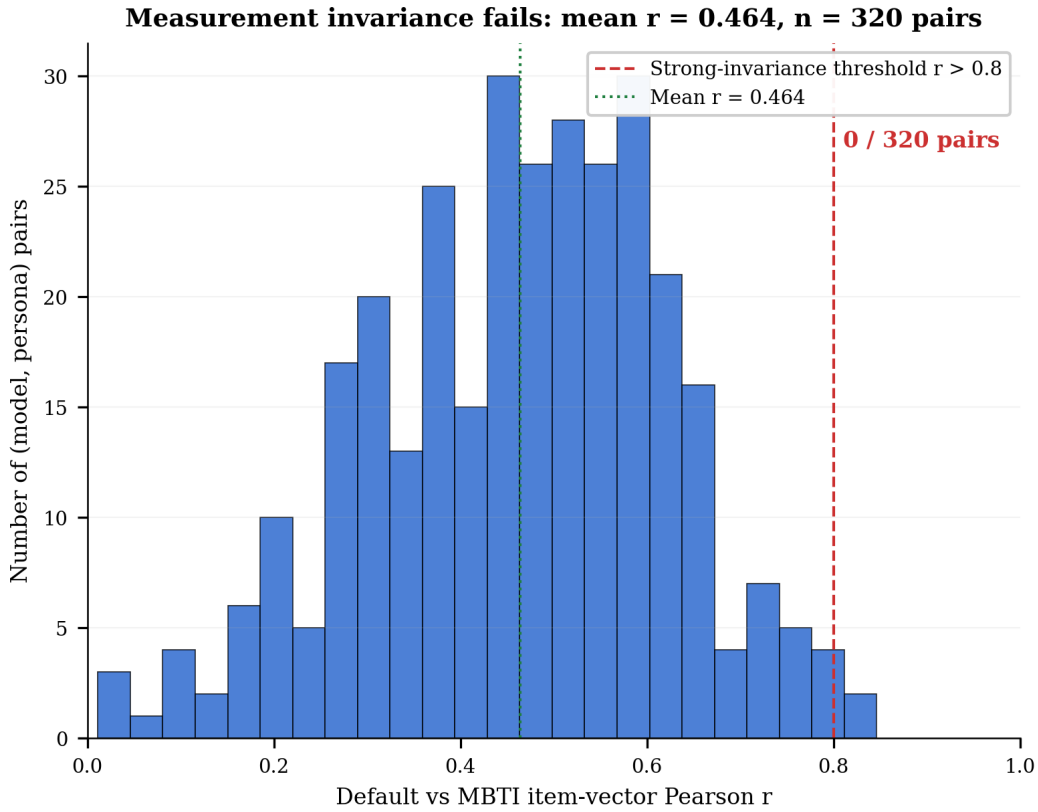


Figure 8: Default-vs-MBTI item-vector Pearson  $r$  across 320 (model, persona) pairs. 0/320 pairs are above the strong-invariance threshold.

#### A.8 PIR BREAKDOWN

Figure 15 compares Default vs MBTI-mean PIR (left) and reports per-model PIR (right).

#### A.9 PER-PERSONA CRONBACH’S $\alpha$

Figure 16 shows per-persona  $\alpha$  on each IPIP Big Five domain. The reverse-density gradient is preserved under every persona.

### B DIFFERENTIAL ITEM FUNCTIONING ACROSS VENDOR FAMILIES

We tested for DIF across two coarse groupings of vendor families: (i) Western (Anthropic, OpenAI, Google) vs Chinese (DeepSeek, Alibaba, Zhipu, Moonshot, MiniMax); (ii) within-vendor-family contrasts using the leave-one-family-out approach. No personality domain shows a statistically significant Western-vs-Chinese gap (Welch’s  $t$ , all  $p > 0.16$ ); the largest effect is  $|d| = 0.42$  on Openness. Per-family EFA reruns (one rerun per vendor family) all return exactly 3 Kaiser factors, with first-eigenvalue range 6.71–7.18 across the 8 families.

### C MISSING-DATA HANDLING

61 refusals out of 375,700 raw API calls (0.0162%) were observed, all under the Default condition (Figure 18). By model: Gemini-3.1-Flash-Lite (34), Gemini-3-Flash-Preview (22), Claude-Sonnet-4.6 (5). By refusal strategy: AI self-identification (57%), philosophical hedging (28%), both (15%). By content theme: politically sensitive (34%), self-referential / emotional (33%), sexually explicit

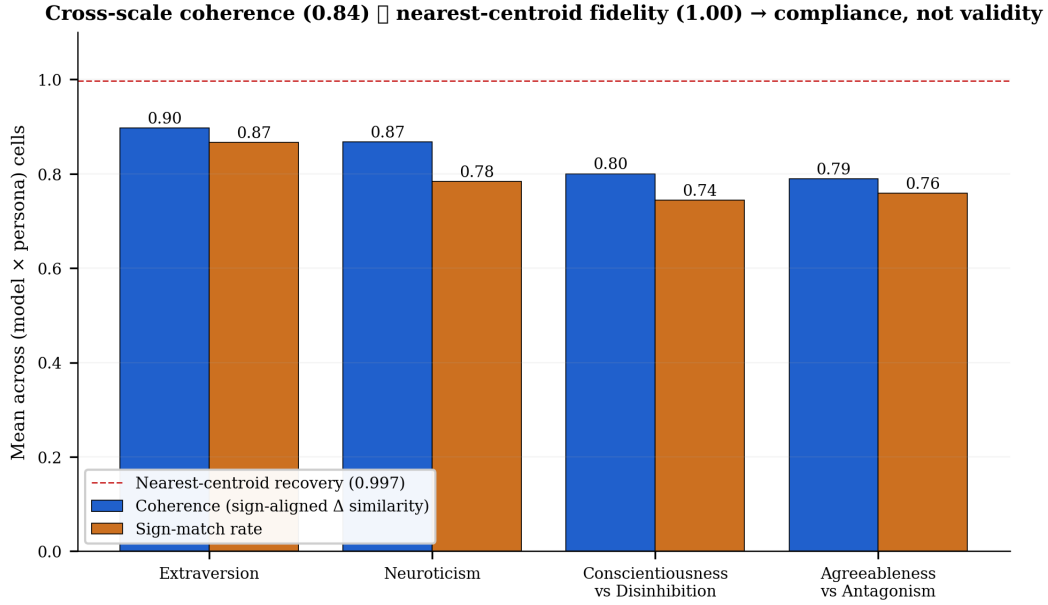


Figure 9: Cross-scale coherence (blue) and sign-match rate (orange) per construct. Far below nearest-centroid fidelity (dashed red).

| Domain                    | F1           | F2           | F3           |
|---------------------------|--------------|--------------|--------------|
| IPIP Neuroticism          | -0.30        | <b>+0.72</b> | +0.39        |
| IPIP Extraversion         | <b>+0.94</b> | +0.03        | +0.29        |
| IPIP Openness             | -0.04        | <b>+0.55</b> | <b>+0.58</b> |
| IPIP Agreeableness        | +0.05        | <b>+0.90</b> | -0.22        |
| IPIP Conscientiousness    | +0.00        | -0.04        | <b>-0.91</b> |
| SD3 Machiavellianism      | -0.09        | <b>-0.82</b> | +0.04        |
| SD3 Narcissism            | <b>+0.91</b> | -0.20        | +0.16        |
| SD3 Psychopathy           | +0.38        | <b>-0.59</b> | <b>+0.63</b> |
| ZKPQ Activity             | <b>+0.90</b> | -0.27        | -0.12        |
| ZKPQ Aggression-Hostility | +0.48        | <b>-0.60</b> | +0.36        |
| ZKPQ Impulsive SS         | +0.34        | +0.08        | <b>+0.84</b> |
| ZKPQ Neuroticism-Anxiety  | -0.27        | <b>+0.83</b> | +0.08        |
| ZKPQ Sociability          | <b>+0.96</b> | +0.04        | +0.13        |
| EPQR Psychoticism         | +0.25        | -0.11        | <b>+0.88</b> |
| EPQR Extraversion         | <b>+0.96</b> | -0.05        | +0.12        |
| EPQR Neuroticism          | -0.04        | <b>+0.91</b> | +0.02        |
| EPQR Lie                  | -0.08        | +0.07        | <b>-0.89</b> |

Table 5: Varimax-rotated factor loadings on the standardised 17-domain matrix ( $n = 340$ ). Bold  $|loading| > 0.5$ . F1 is Extraversion / Sociability; F2 fuses Neuroticism with (inverted) Agreeableness / Machiavellianism; F3 contrasts Conscientiousness with Impulsive Sensation Seeking, Psychoticism, and the Lie scale.

(13%), social-manipulation / danger-seeking (13%). The single most-refused item is SD3-026 (“I enjoy having sex with people I hardly know.”, 8 refusals). Missing cells were filled by per-item mean imputation across remaining models and samples under the same Default condition; the rate is far below any threshold at which the imputation choice could materially affect downstream statistics.

## D WITHIN-SAMPLE RESPONSE CONSISTENCY

Figure 19 reports per-model within-item SD across the  $k = 5$  repeated samples on the Likert composite. The total range spans an order of magnitude: Claude-Opus-4.6 (0.019, near-deterministic)

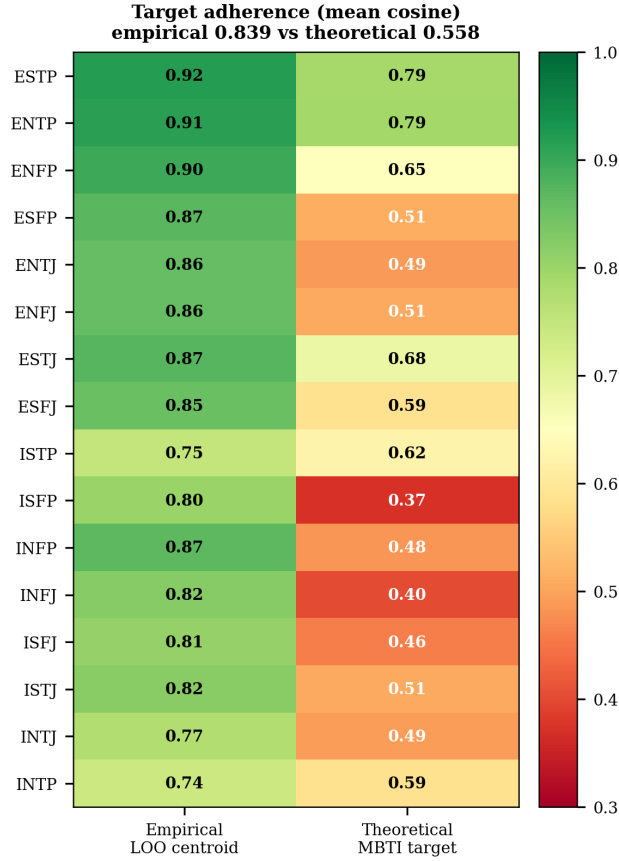


Figure 10: Per-persona target adherence: LOO-empirical (left column) vs theoretical MBTI mapping (right column).

| Pair (across 20 models)                 | Sign | $r_p$ | $p_p$ | OK?       |
|-----------------------------------------|------|-------|-------|-----------|
| Neuroticism $\times$ N-Anxiety          | +    | 0.64  | 0.002 | yes       |
| Extraversion $\times$ Sociability       | +    | -0.10 | 0.670 | <b>NO</b> |
| Extraversion $\times$ EPQR Extraversion | +    | 0.51  | 0.020 | yes       |
| Neuroticism $\times$ EPQR Neuroticism   | +    | 0.48  | 0.031 | yes       |
| Agreeableness $\times$ Machiavellianism | -    | -0.71 | 0.000 | yes       |
| Agreeableness $\times$ Psychopathy      | -    | -0.81 | 0.000 | yes       |
| Conscientiousness $\times$ Psychopathy  | -    | -0.73 | 0.000 | yes       |
| EPQR Psychoticism $\times$ Psychopathy  | +    | 0.11  | 0.646 | yes       |

Table 6: Pearson convergent-validity correlations across the 20 model-level means for 8 pre-specified construct pairs. Sign-match 7/8, with the single sign reversal on Extraversion  $\times$  ZKPQ Sociability.

to DeepSeek-V4-Pro (0.192). What looks like a personality difference between Claude and DeepSeek at the profile level is in part a sampling-noise difference. This motivates the multi-seed / multi-temperature recommendation in the Conclusion and is itself a reason to be cautious about cross-model trait comparisons.

**Per-persona stability.** Mean within-item SD per persona ranges from 0.081 (ISFJ) to 0.118 (ISTP); the Default condition is the noisiest of all at 0.205. Persona prompts therefore reduce response stochasticity rather than increasing it — giving the model a coherent character to inhabit reduces the alignment-vs-honesty tension that drives Default-condition jitter.

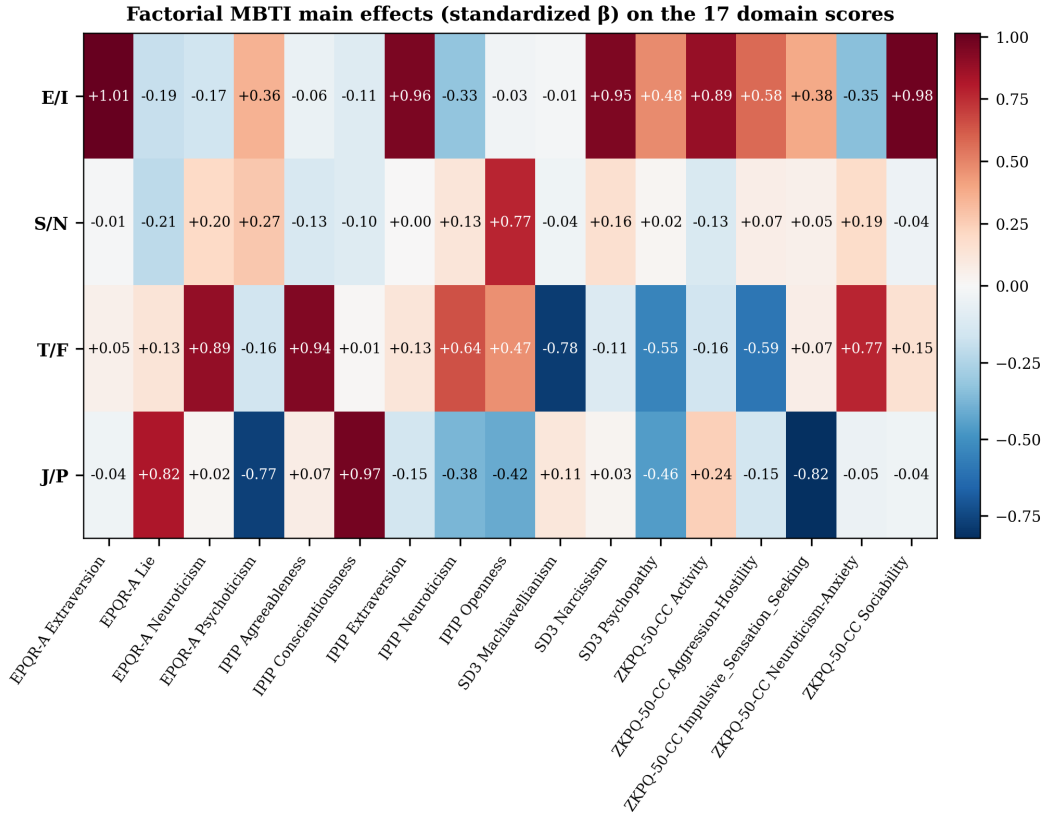


Figure 11: Standardised main-effect coefficients of the four MBTI axes on the 17 domain scores.

| Strategy           | N           | E           | O           | A           | C           |
|--------------------|-------------|-------------|-------------|-------------|-------------|
| Random             | 0.08        | -0.01       | 0.06        | 0.04        | -0.11       |
| Pure Acquiescence  | 0.02        | 0.03        | -0.01       | 0.10        | -0.03       |
| Trait+Acquiescence | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> |
| LLM observed       | 0.81        | 0.93        | 0.19        | 0.07        | 0.54        |
| Human (Johnson)    | 0.90        | 0.89        | 0.87        | 0.88        | 0.90        |

Table 7: Cronbach’s  $\alpha$  per IPIP Big Five domain for three synthetic baselines vs LLM observed and human normative sample.

**Binary scales.** On binary scales the picture is sharper: mean within-item SD is 0.037 on ZKPQ true/false and 0.032 on EPQR yes/no, with mode-agreement rates above 0.99 on most (model, persona) cells.

## E SAMPLE-COUNT CONVERGENCE

To confirm that  $k=5$  samples per cell suffice, we recompute the three headline diagnostics using only the first  $k \in \{1, 2, 3, 4, 5\}$  samples and correlate each against the  $k=5$  baseline across the 20 models (Spearman  $\rho$ ). For Cronbach’s  $\alpha$ , four of the five IPIP domains exceed  $\rho=0.98$  by  $k=3$ ; Openness, the most reverse-heavy domain (38%), reaches  $\rho=0.94$  at  $k=3$  and 0.97 at  $k=4$ . For the Pairwise Inconsistency Rate,  $\rho(k=3, k=5) = 0.98$  (mean absolute error 0.014; 0.008 at  $k=4$ ). The EFA factor structure is perfectly stable:  $k=1, 3, 5$  each yield exactly three factors under both the Kaiser criterion and parallel analysis, with near-identical eigenvalues (top three 6.84/4.33/3.00, 6.96/4.32/3.06, 6.97/4.33/3.07).  $k=3$  is therefore sufficient and  $k=5$  conservative; no main conclusion depends on the sample count.

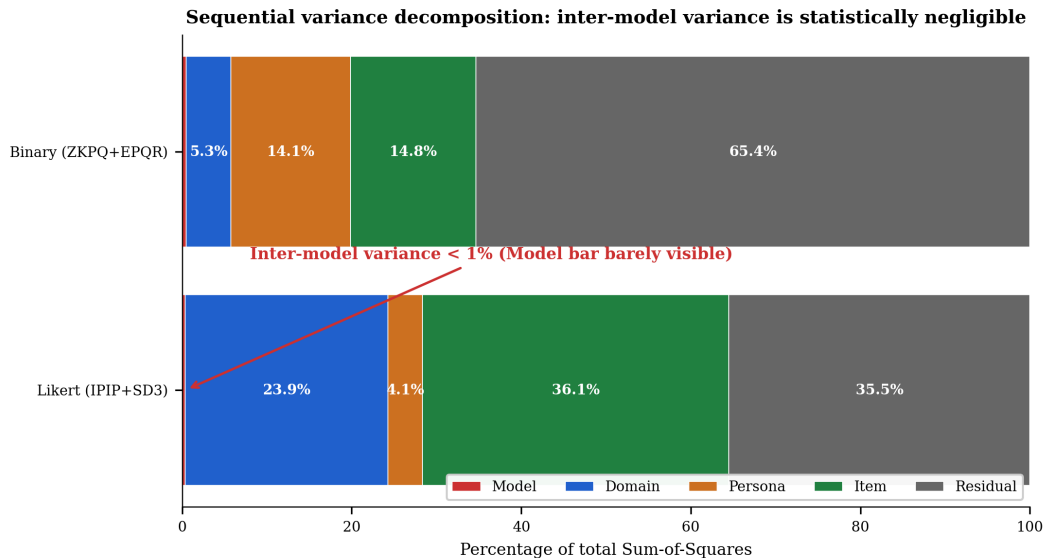


Figure 12: Sequential SS decomposition into Model / Domain / Persona / Item / Residual for the Likert (top) and Binary (bottom) composites. The Model slice is barely visible (0.35% Likert; 0.51% Binary) — cross-model personality comparisons are statistically negligible.

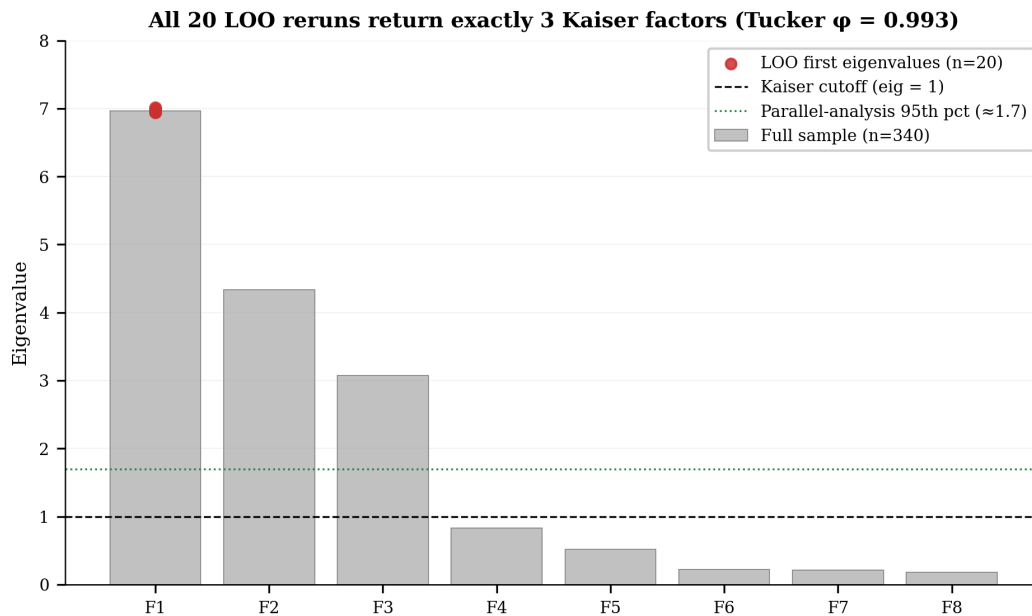


Figure 13: Leave-one-model-out replication of the 17-domain EFA. All 20 reruns return exactly 3 Kaiser factors (red dots at  $F_1$ ).

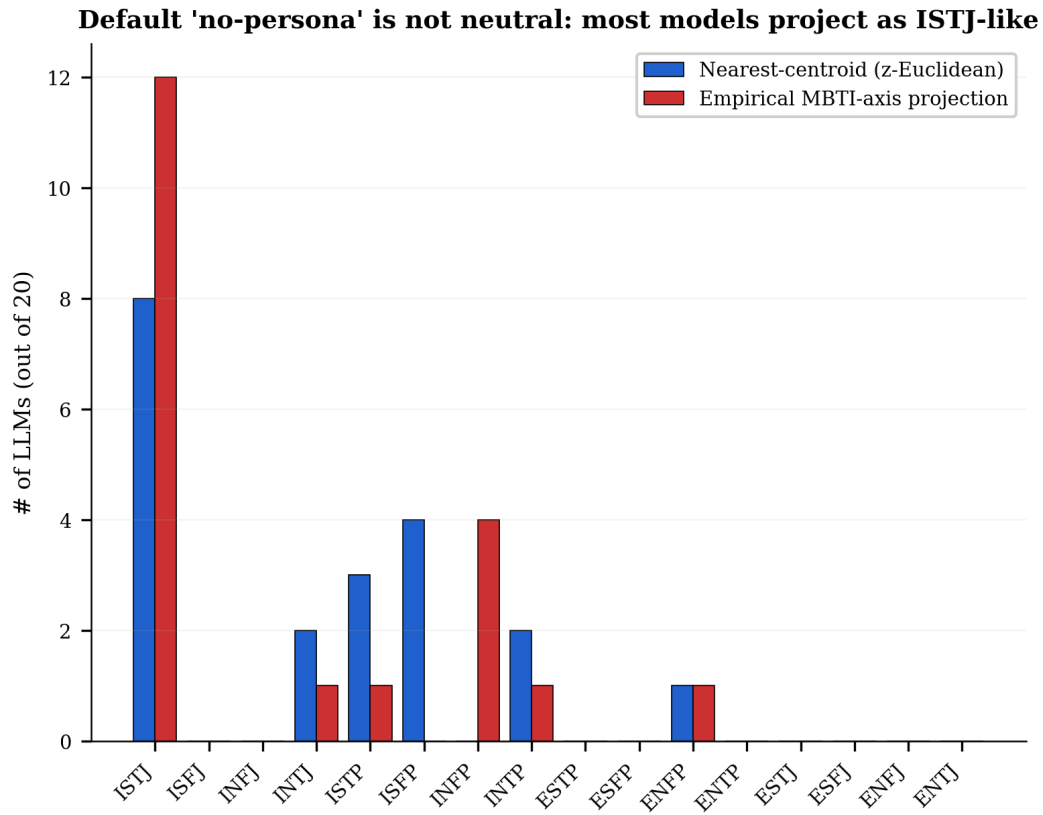


Figure 14: Default-condition profile classification: nearest-centroid (blue) and empirical MBTI-axis projection (red). 12/20 LLMs project as ISTJ-like.

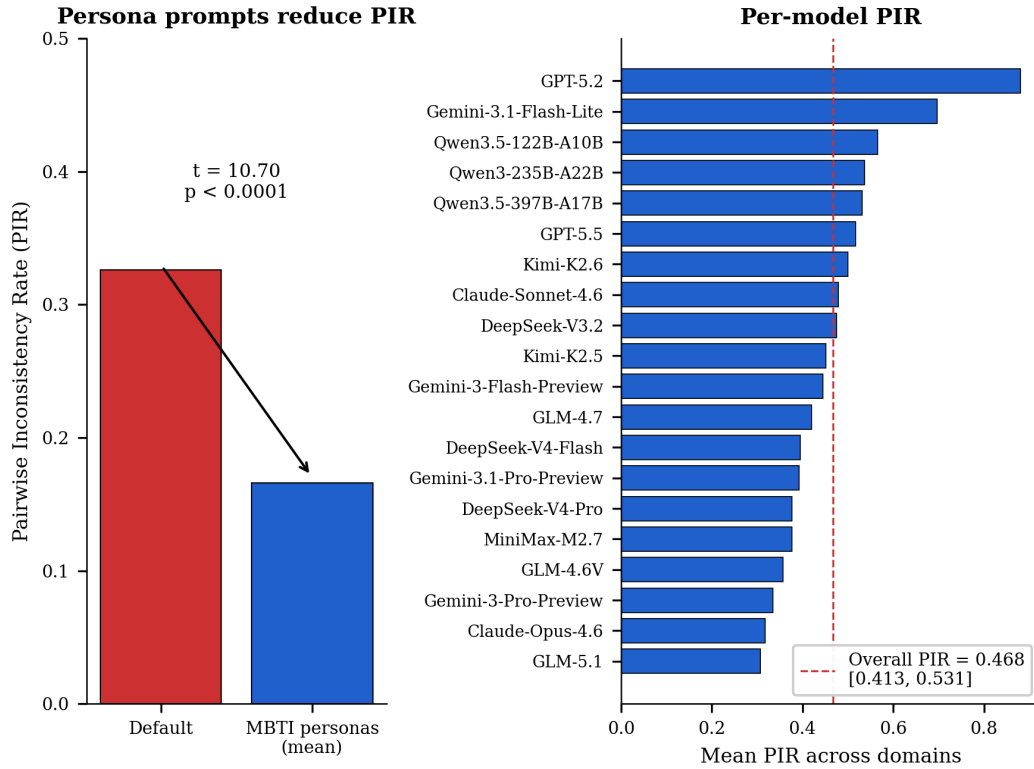


Figure 15: Left: Default vs MBTI-mean Pairwise Inconsistency Rate (paired  $t = 10.70$ ,  $p < 0.0001$ ). Right: per-model PIR with overall PIR = 0.468.

Per-persona  $\alpha$  on IPIP Big Five: low-reverse domains spuriously high, Agreeableness collapses

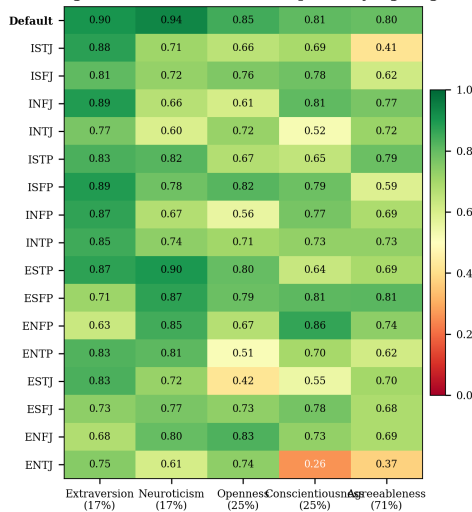


Figure 16: Per-persona Cronbach's  $\alpha$  on IPIP Big Five (columns ordered by reverse-item percentage). Default row in bold. The reverse-density gradient is invariant under role-playing.

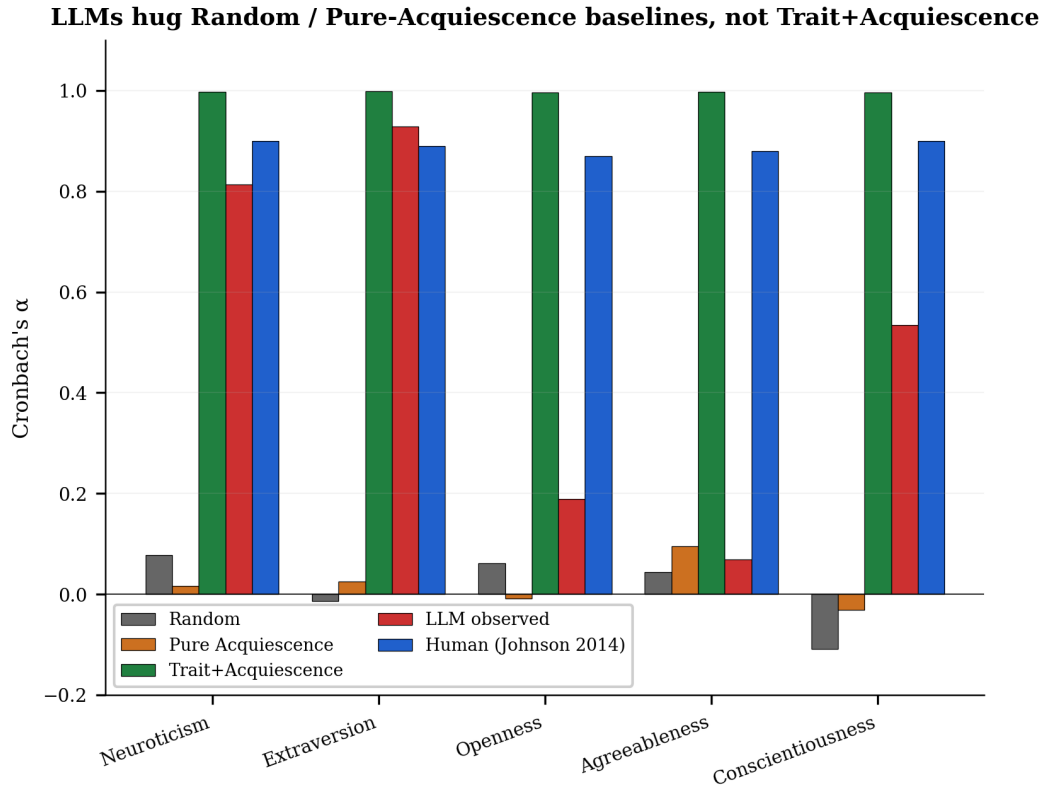


Figure 17: Cronbach’s  $\alpha$  per IPIP Big Five domain for Random, Pure Acquiescence, Trait+Acquiescence, LLM observed, and the human Johnson (2014) baseline.

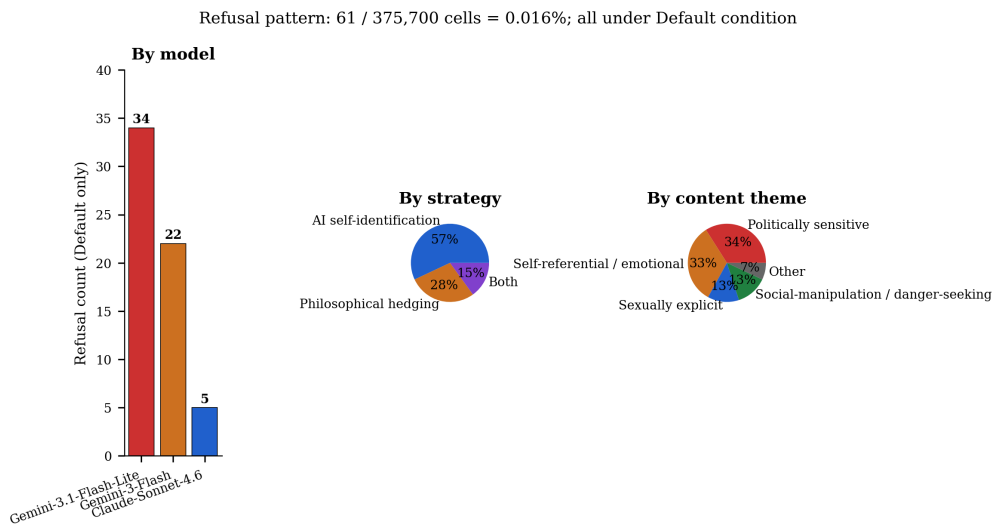


Figure 18: Refusal pattern: 61 / 375,700 cells = 0.0162%; all under Default. Left: counts by model. Center: refusal strategy. Right: content theme.

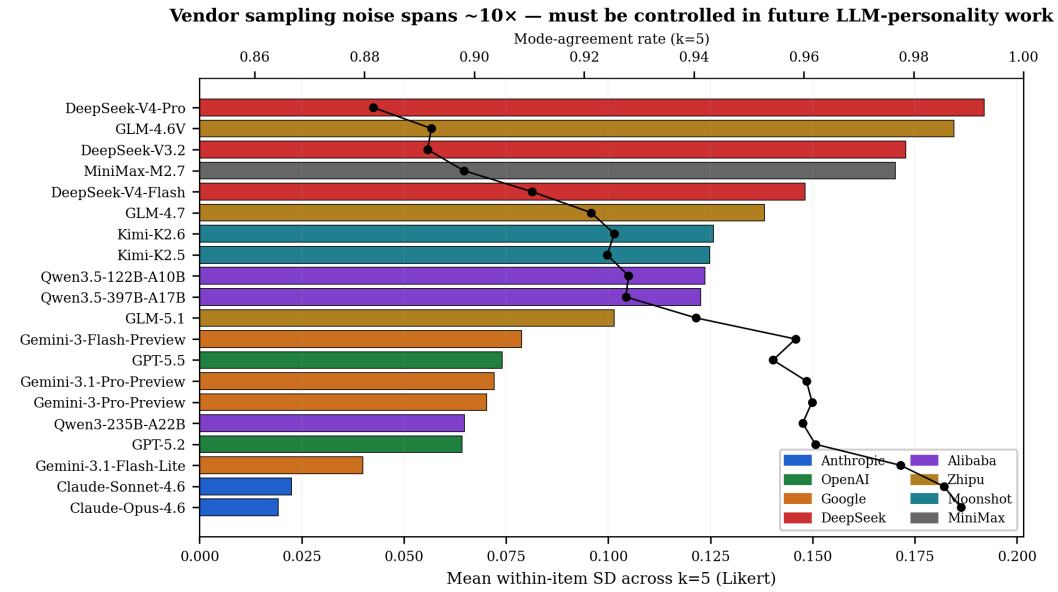


Figure 19: Within-sample stability across  $k = 5$  repeats per model (Likert items). Bars: mean within-item SD coloured by vendor family. Line: mode-agreement rate on the secondary axis. Vendor sampling noise spans  $\approx 10\times$ .